

E7 Introduction to Computer Programming for Scientists and Engineers

Lecture week 15-B

Randomness - continued

Calculating the probabilities of events from distributions

- X is a random variable with sample space Ω and distribution p .
- An **event** of X is any subset of its sample space: $e \subseteq \Omega$
- The probability that event e occurs is the integral (or sum) of $p(x)$ over e .

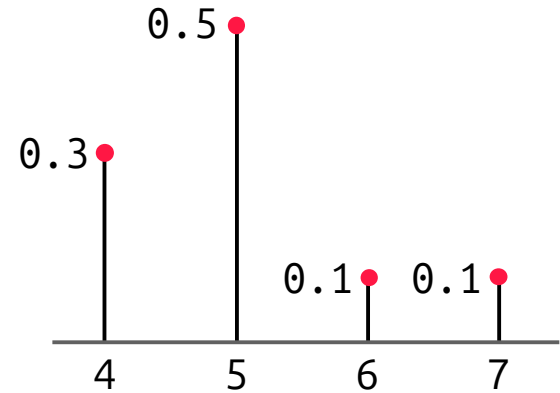
Continuous r.v.

$$\text{Prob}(e) = \int_e p(x) dx$$

Discrete r.v.

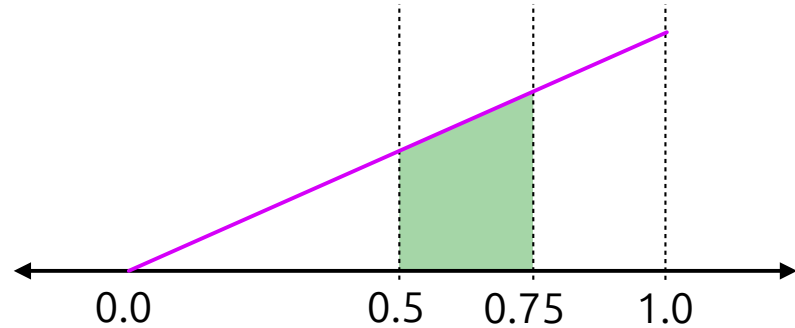
$$\text{Prob}(e) = \sum_{x \in e} p(x)$$

Example For a discrete r.v. with distribution shown below, what is the probability of the event $\{4,6\}$?



Example

For the continuous random variable with distribution show below, what is the probability of the event $[0.5, 0.75]$?



Estimating the probabilities of events from samples

Given a dataset $\mathcal{D}$ consisting of n samples of X

$$\mathcal{D} = \{x_1, \dots, x_n\}$$

we can estimate the probability of an event e with

$$\hat{P}(e) = \frac{\# \text{ samples in } e}{n}$$

Example

Given the dataset plotted below, estimate the probability of the event $[0.5, 0.75]$

